

Data Visualization

Syllabus Discussion

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Assistant Professor of Finance, Economics & Analytics,
Baldwin Wallace University

Agenda

1. General info

- Instructor
- Course
- Basic roadmap

2. Basics and Introduction

General info

Instructor

Instructor Profile

- *Name:* Suresh L. Paul
(<https://www.linkedin.com/in/sureshlpaul/>)
- *Teaching interests:* economics, analytics, ai
- *Research interests:* economics (applied), ai
- *Specific research topics:*
 - Economics of AI
 - Corporate Governance.— CEO pay and related studies
 - Health economics.— opioids, disease transmissions
 - Labor economics.— gender wage/participation gap

Instructor Profile (cont'd.)

- *Qualification/Degrees Earned:*
 - CPhil Finance, **Binghamton University**
 - MS Industrial Engg (Robotics), **University of Cincinnati**
 - BS Mechanical Engg, **CEG, Anna University, India**
- *Currently:*
 - Founder & Research Scientist, **Algorithm Basics**
 - Consultant, **Precision Compounds Inc**
 - Professor of Finance & Analytics, **Baldwin Wallace Univ**

General info

Course

Course info

- BUS623
- DATA VISUALIZATION
 - ... FOR MANAGERS
 - *using R and AI assistants*

Course info

- Working knowledge in
 - *R*
 - Visualization
 - *Dashboards*
 - Shiny App
 - Implementation using AI Applications

Meeting info

- *Meeting room*: Bloomberg Research Center
- *Timing*: Mondays 6:15PM - 9:00PM
- *Office hours*: by appointment only
- *Textbook*: None.
 - all codes and materials will be made available via Canvas.
- *Computing requirements*: Lab / Personal Laptops.
 - R / RStudio
 - appointment with IT for personal computing needs.
- *AI tools*:
 - any code-producing assistant (Claude / ChatGPT / Gemini); free tier

Why AI assistants?

BIG IDEAS 2023: ARTIFICIAL INTELLIGENCE

27



AI Could Lead To A 10-Fold Increase In Coding Productivity

Based on a 70% annualized drop in training costs and feedback loops, AI coding assistants like Copilot could increase the output of software engineers ~10-fold by 2030.

Github Copilot Example

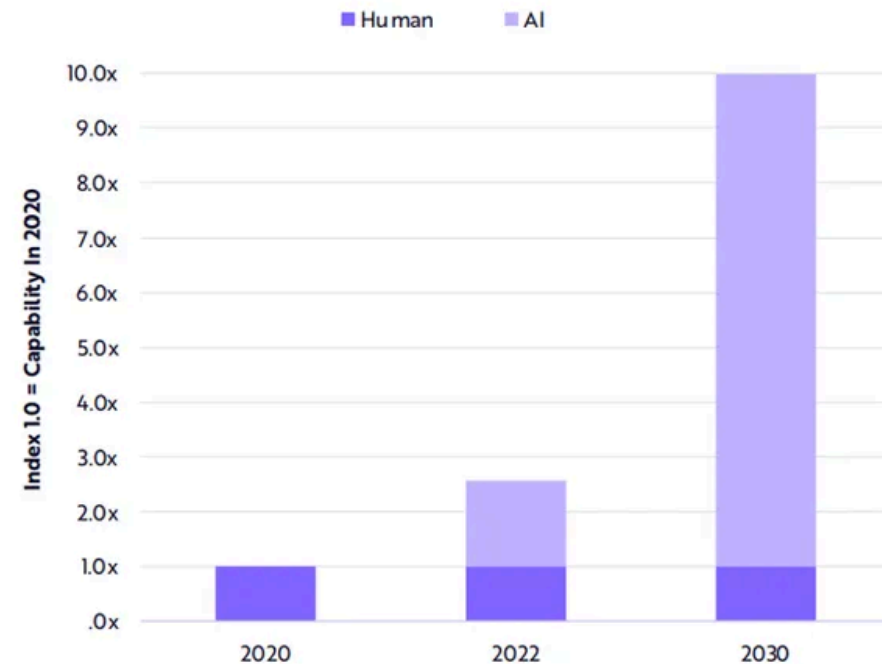
Human Input

```

1 #!/usr/bin/env ts-node
2
3 import { fetch } from "fetch-h2";
4
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8   const response = await fetch('http://text-processing.com/api/sentiment/', {
9     method: "POST",
10    body: `text=${text}`,
11    headers: {
12      "Content-Type": "application/x-www-form-urlencoded",
13    },
14  });
15  const json = await response.json();
16  return json.label === "pos";
17 }
  
```

Copilot

Output of Human + AI: Coding Tasks



Sources: ARK Investment Management LLC, 2023. GitHub 2023. Forecasts are inherently limited and cannot be relied upon. For informational purposes only and should not be considered investment advice or a recommendation to buy, sell, or hold any particular security. Past performance is not indicative of future results.

General info

roadmap

Roadmap

Foundations — Classes 1–3 (*AI-OFF*)

- *Class 1*: Intro, Concepts, Data Types & Structures, Visual Literacy
- *Class 2*: Grammar of Graphics; base *R* plots (as contrast); intro to *ggplot2*
- *Class 3*: *tidyverse* & *dplyr*; core plot types with *ggplot2*

Roadmap (cont'd.)

Applied — Classes 4–6 (*AI-assisted* → *AI-encouraged*)

- *Class 4*: Choosing the right chart; anti-patterns; chart-lie detection; accessibility
- *Class 5*: Interactivity & Dashboards (*Shiny*)
- *Class 6*: Storytelling & Conceptualization

Roadmap (cont'd.)

Project — Classes 7–8 (*AI-encouraged*)

- *Class 7*: Project self-work (instructor available for consultation)
- *Class 8*: Final Project presentations & live defense

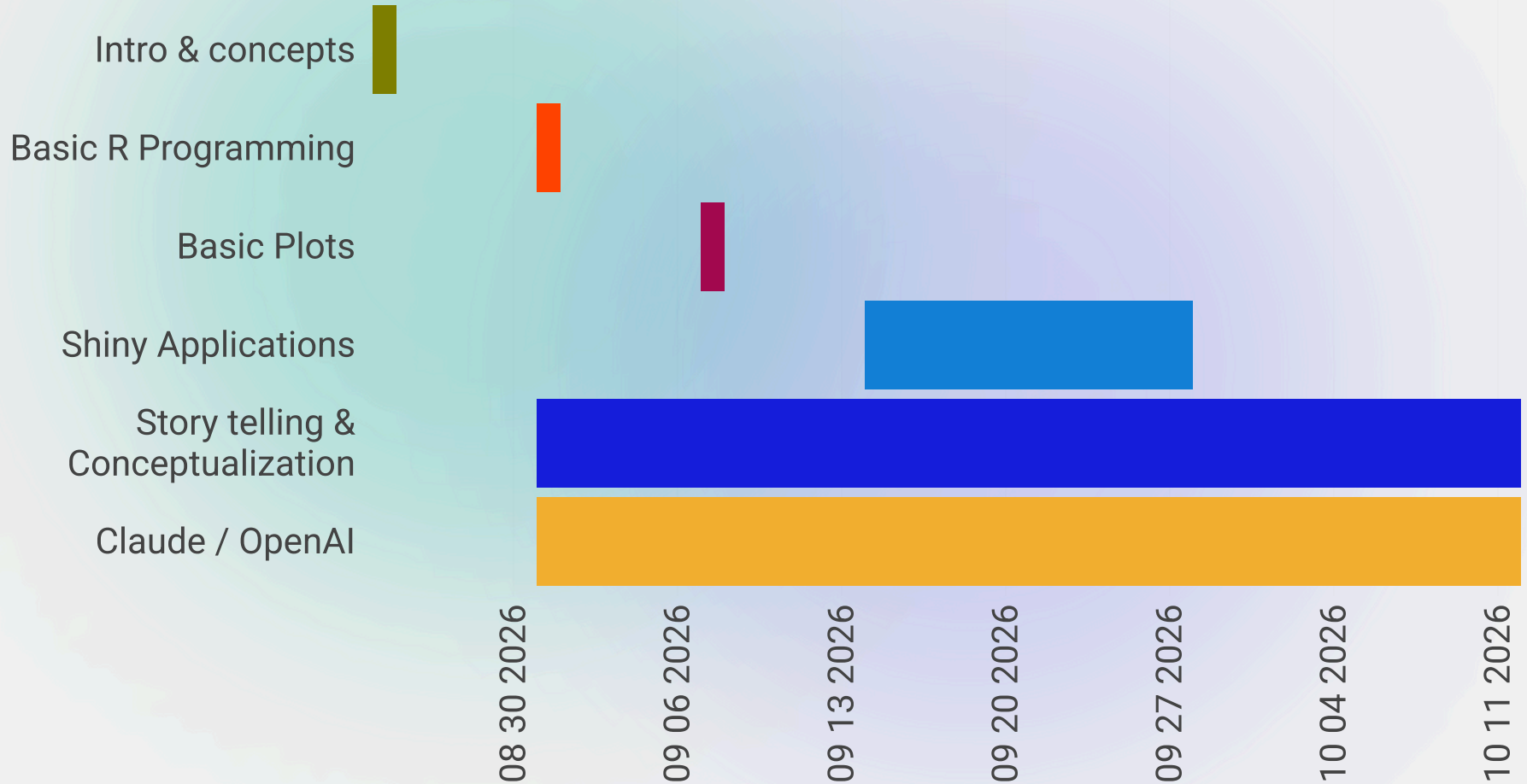


Introduction to

Data Visualizations

Suresh L. Paul

Topic Schedule



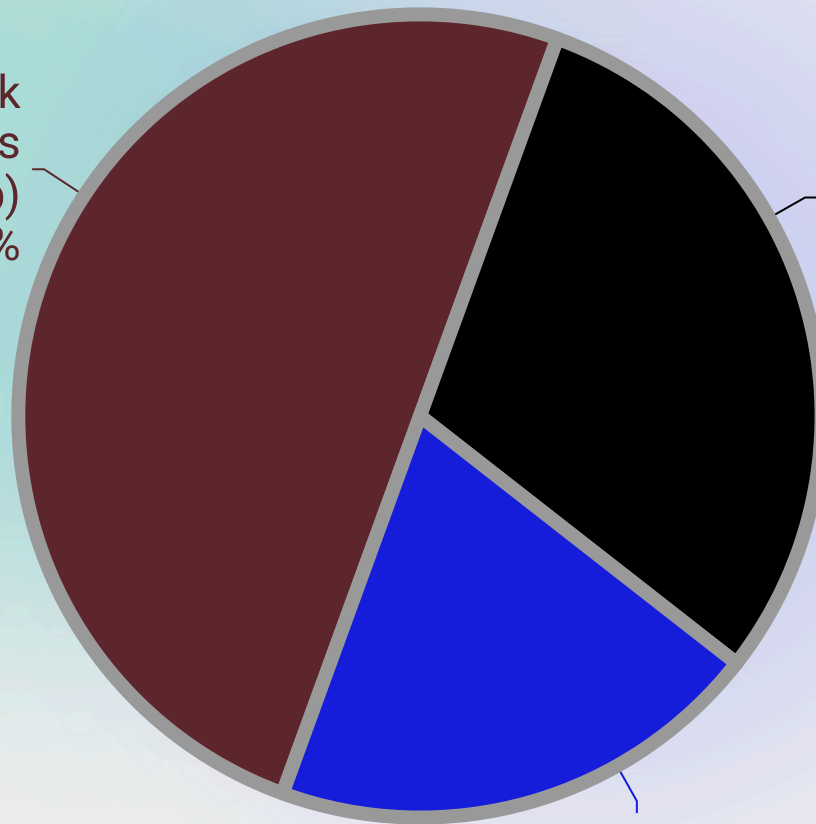
Grading Scheme



Homework
Assignments
(group)
50.00%

Project
Presentations
(individual)
30.00%

Class Participation (individual)
20.00%



Tools & Materials



Power BI

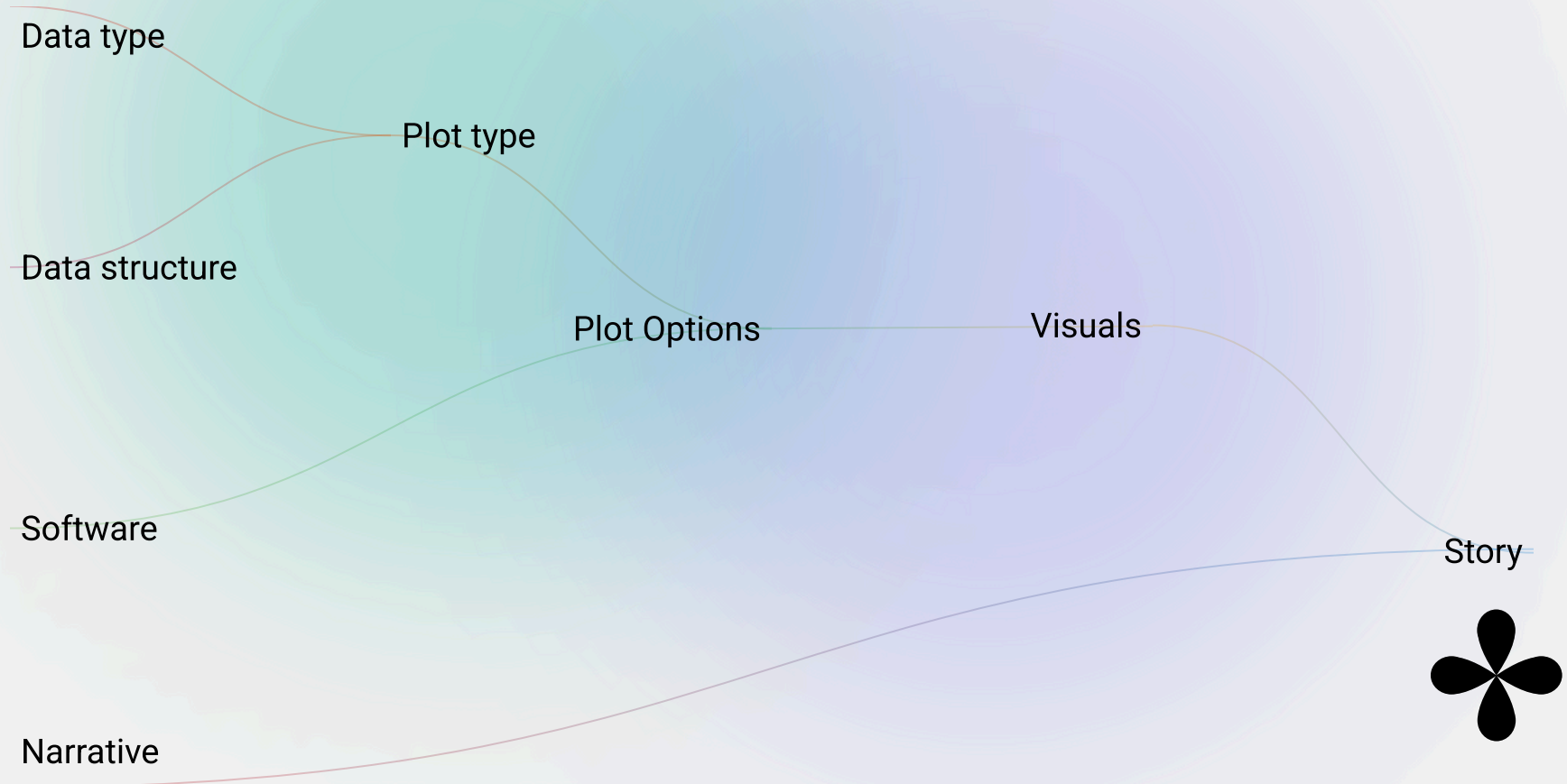
RStudio

OpenAI Claude

Canvas

MS Excel Datasets

Topic 1: Intro/Concepts





- Categorical data - Nominal, Ordinal, Binary, Boolean
- Numerical data - Discrete, Continuous
- Time-based data - Time Series, Cross-sectional, Panel
- Spatial data - Point, Polygon, Raster

Data type

Plot type

Data structure

- Structured data - Lists, Arrays, DataFrames, Spreadsheet
- Unstructured data - Free text, HTML, XML, JSON
- Specialized data - Hierarchical, Network, Image, Audio

Treemap
Histogram
Area Bubble
Stacked bar
Gantt Venn
Map
Line
Box
Sankey
Funnel
Scatter
Sunburst
Bar
Heatmap
Pie
Radar
Donut
Stacked Area
Waterfall
Candlestick

Plot Options

Plot type

Treemap
Histogram
Area Bubble

Stacked bar

Gantt Venn
Line

Map
Box
Funnel
Scatter
Sunburst
Bar
Sankey
Heatmap
Pie
Radar

Donut
Stacked Area
Waterfall
Candlestick

Plot Options

Animation Dynamic
Dashboard
WebApp
Interactive
Static

Data structure

Software

MS Excel
Tableau
Minitab
Julia
SPSS
Stata
Power BI
SQL
SAS
R
Python



Plot Options

Animation

Visuals

Interactive

Dashboard WebApp

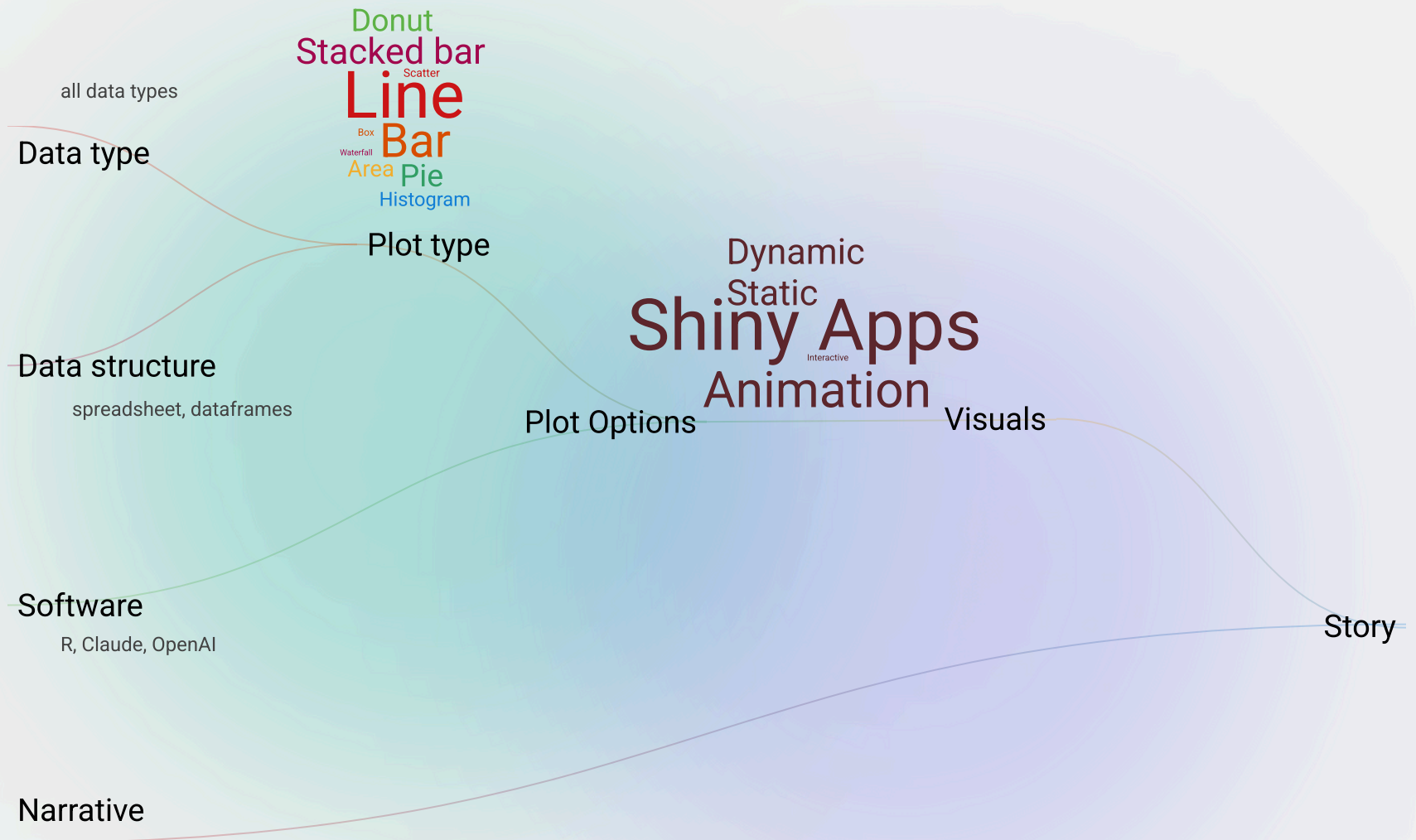
Static
Dynamic

Story

Narrative

Knowledge
Context
Perspective

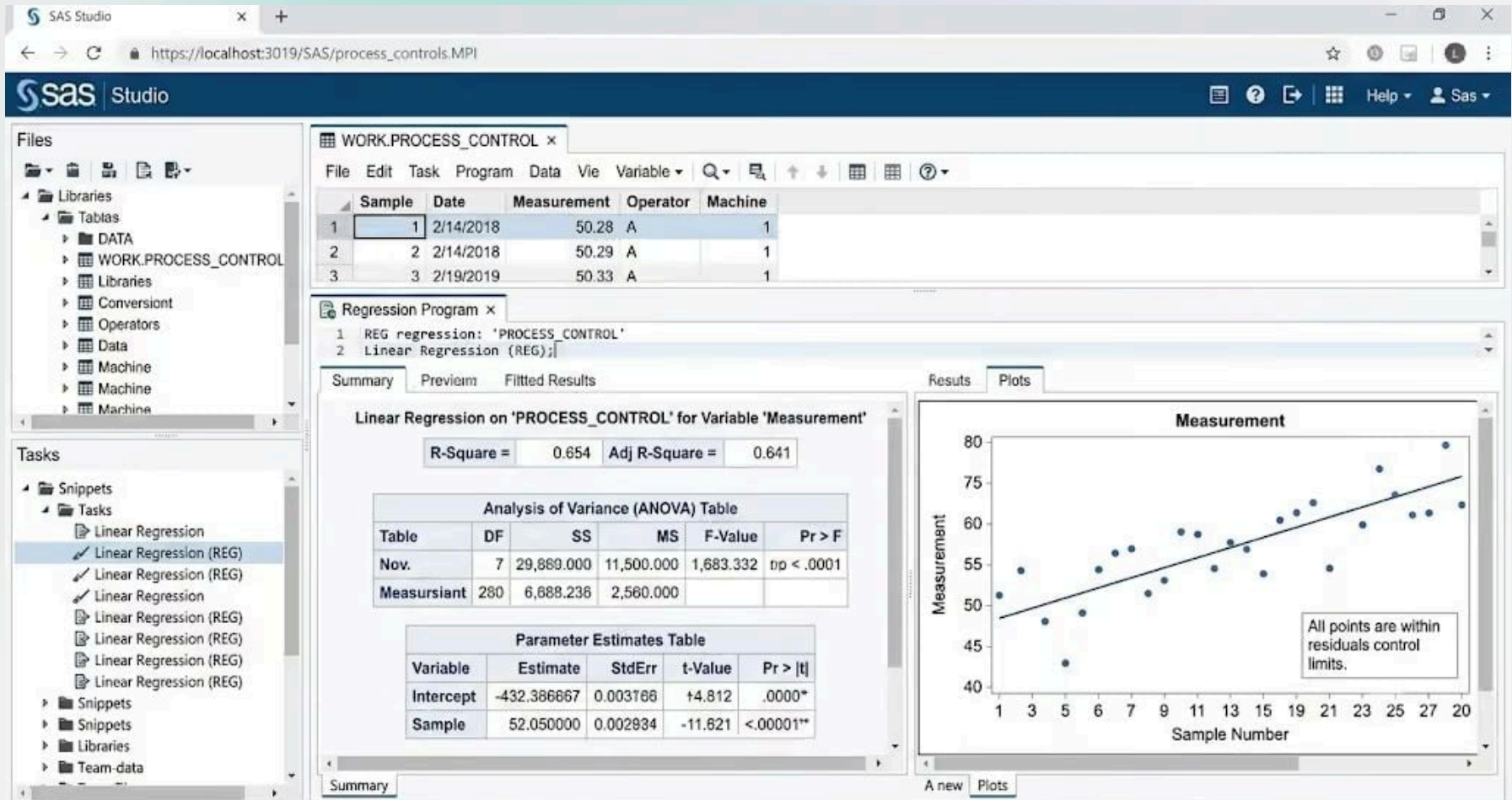




... in this course

Why R?

Industry Standard Computing

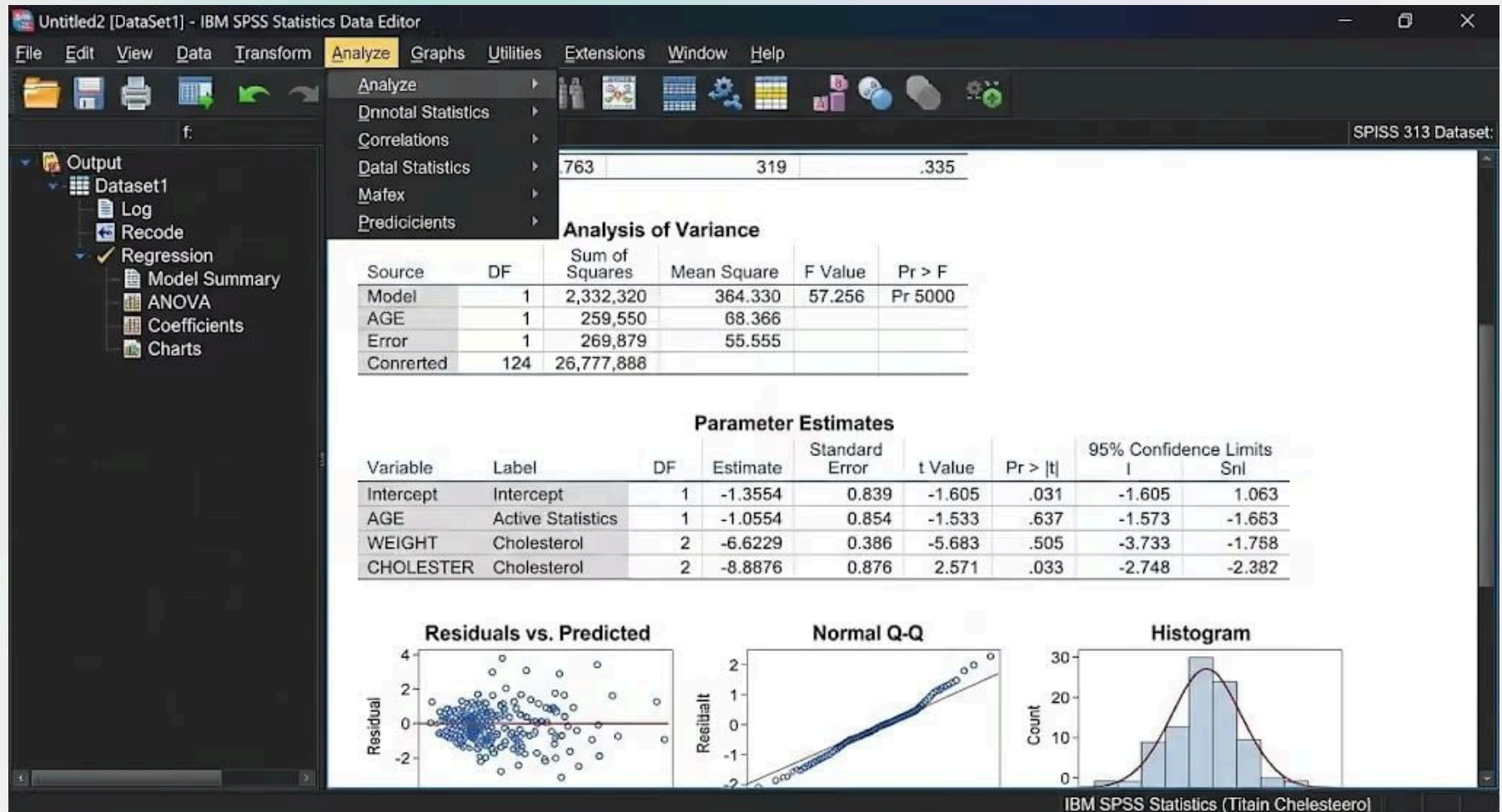


SAS Studio - analytical workflow - linear regression complete.

Why R?



SPSS

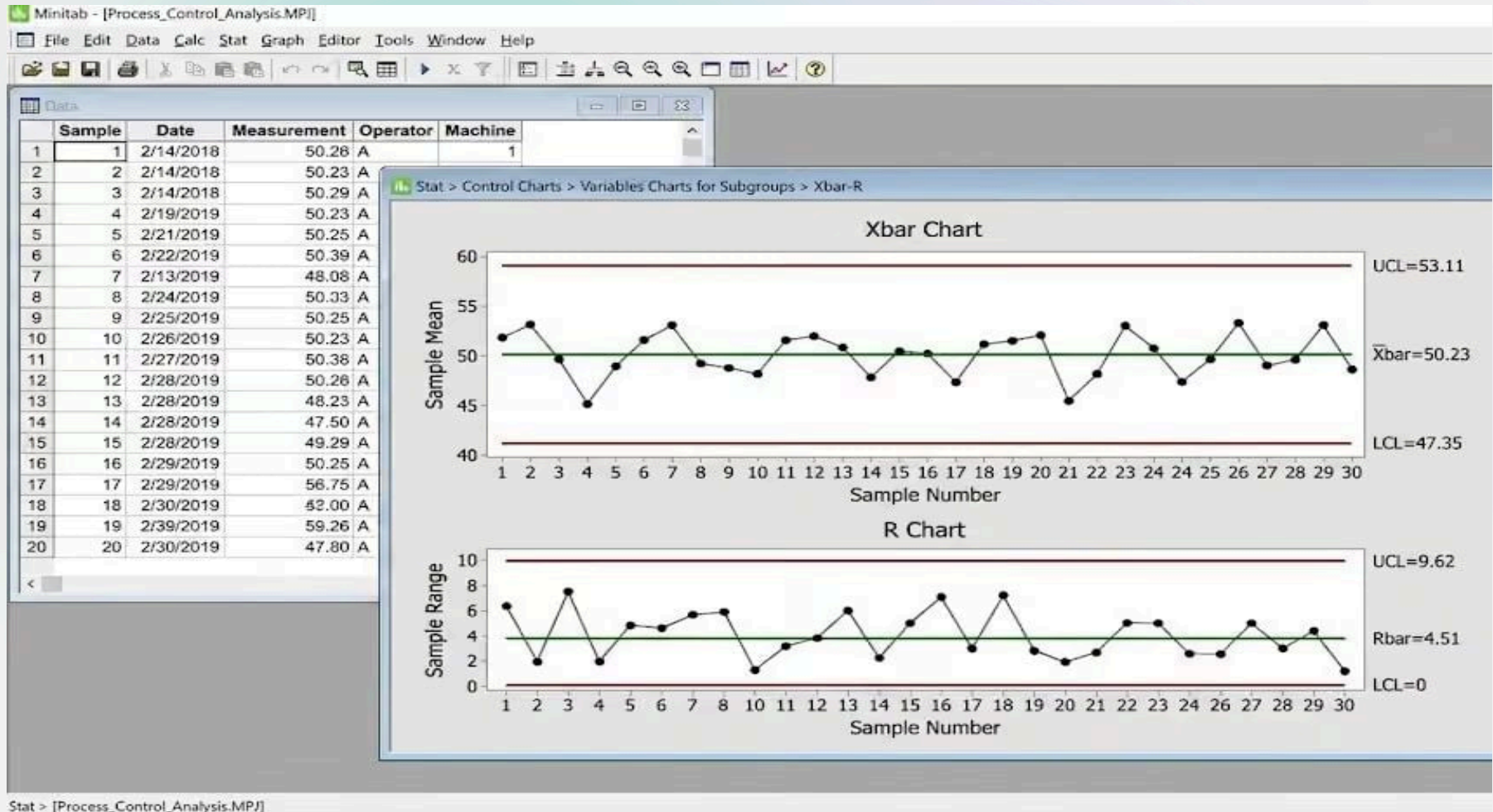


Why R?



```
Stata/IC 17.0 - running
File Edit View Options Window Help
Variables
Name
wages
education
experience
training
gender
prox_train_ctr
Results
. import excel "fictional_labor_data.xlsx", sheet("TrainingImpact") firstrow
(6 variables, 1,024 observations)
. describe
Contains data from fictional_labor_data.dta
Observations: 1,024
Variables: 6
Fictional Labor Data
Training Impact Study
Variable name Storage type Display format Value label Variable label
wages double %10.0g Annual Wages (thousands of USD)
education byte %10.0g Years of Schooling
experience byte %10.0g Years of Work Experience
training byte %10.0g Participated in Training Program (1=Yes, 0=No)
gender byte %10.0g Gender (1=Female, 0=Male)
prox_train_ctr double %10.0g Proximity to Training Center (miles)
. sum wages education experience training gender prox_train_ctr
Variable | Obs Mean Std. dev. Min Max
wages | 1,024 68.4523 18.3217 25 150
education | 1,024 13.784 2.103 8 20
experience | 1,024 11.129 7.452 0 40
training | 1,024 0.392 0.488 0 1
gender | 1,024 0.471 0.499 0 1
prox_train_ctr | 1,024 5.1091 3.4567 0.1 18.5
. ivregress 2sls wages experience gender (training = education prox_train_ctr)
Instrumental variables (2SLS) regression
Number of obs = 1,024
Wald chi2(3) = 84.15
Prob > chi2 = 0.0000
R-squared = -0.1251
Koot MSE = 18.983
wages | Coefficient Std. err. z P>|z| [95% conf. interval]
training | 12.4567 3.5678 3.49 0.000 5.4632 19.4502
experience | 0.8712 0.0984 8.85 0.000 0.6782 1.0641
gender | -5.1234 1.2345 -4.15 0.000 -7.5432 -2.7036
_cons | 38.9876 4.1234 9.45 0.000 30.9056 47.0696
Instrumented: training
Instruments: experience gender education prox_train_ctr
. estat endogenous
Tests of endogeneity
Ho: variables are exogenous
Durbin (score) chi2(1) = 15.845 (p = 0.0001)
Wu-Hausman F(1,1021) = 15.912 (p = 0.0001)
Command
import excel "fictional_labor_data.xlsx", sheet("TrainingImpact") firstrow
describe
sum wages education experience training gender prox_train_ctr
ivregress 2sls wages experience gender (training = education prox_train_ctr)
estat endogenous
```

Why R?

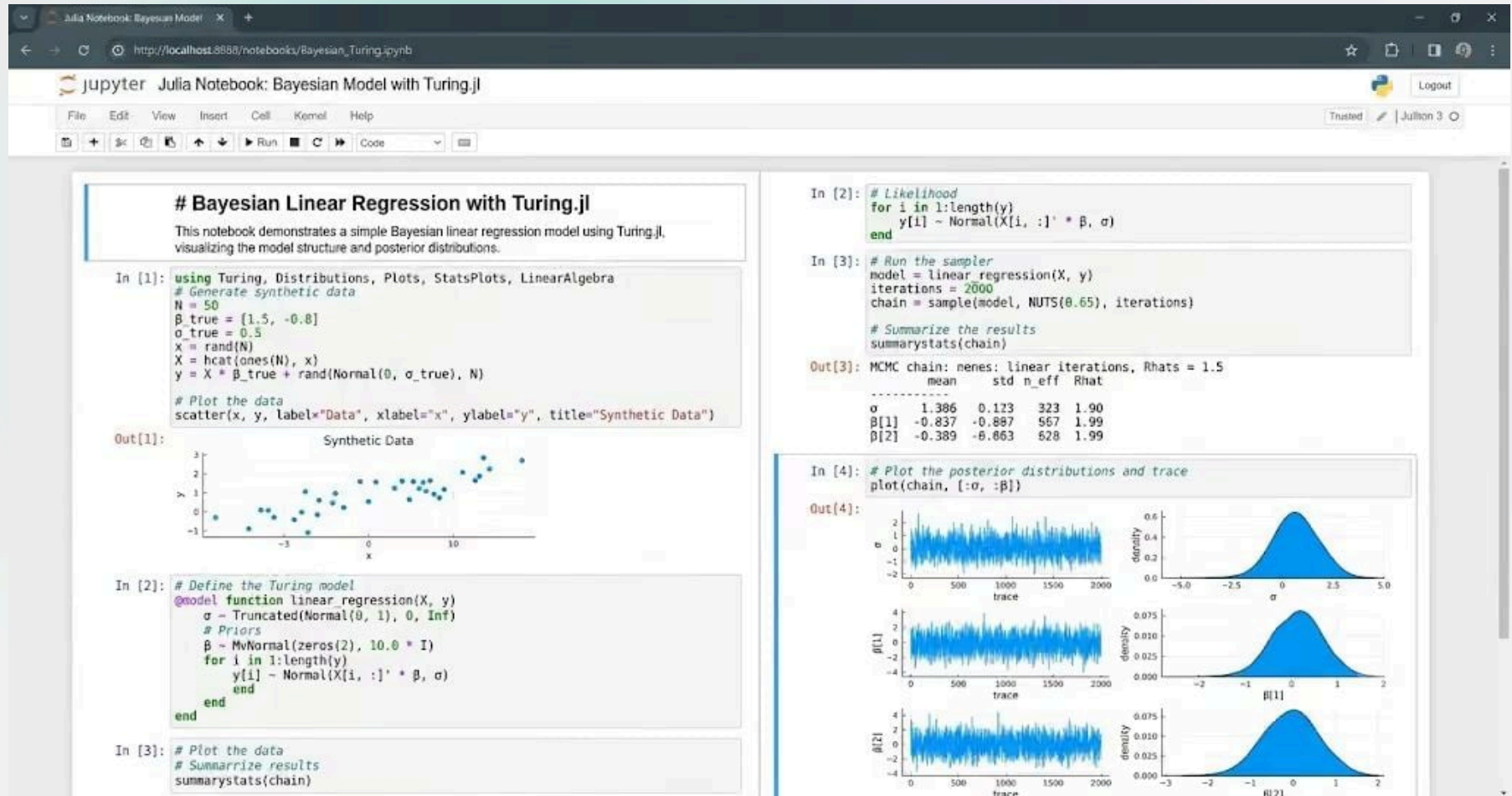


Why R?

High-Performance Statistical Computing



Julia

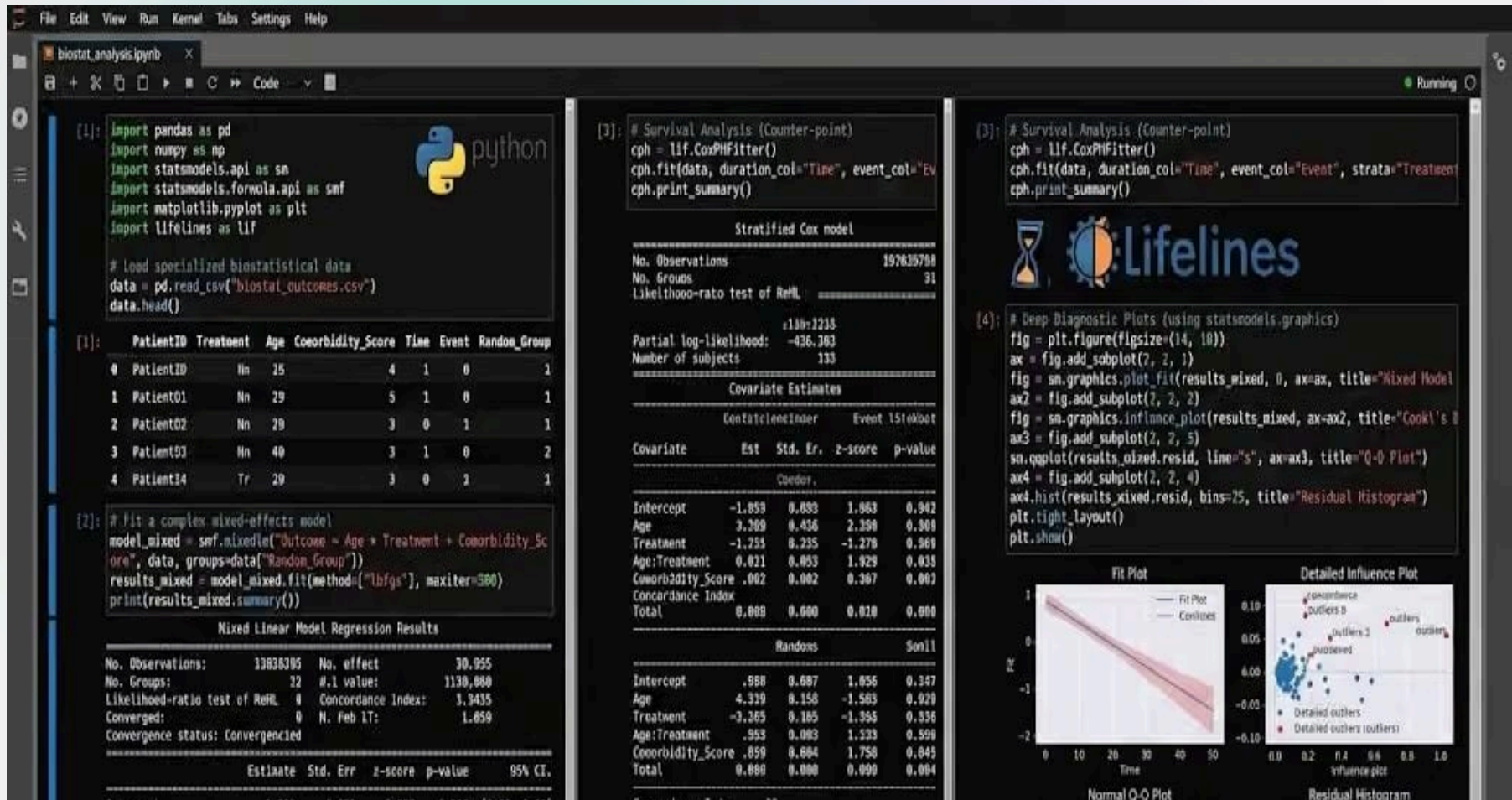


Why R?

General-Purpose Statistical Work

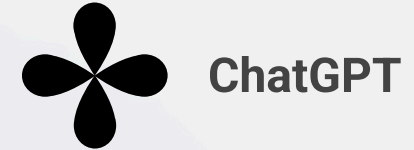


Python



Why R?

General-Purpose Statistical Work



OpenAI - ChatGPT

Recent Chats

Manufacturing Data Hypothesis Test (Julius)

Manufacturing Data Hypoth...

Recent chats tested

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Process_Control_Analysis.M...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Manufacturing Data Hypoth...

Recent Chats



Julius, can you run a t-test to see if the Measurement values for Operator A are significantly different from Operator B? Use the uploaded manufacturing data (Process_Control_Analysis.MPJ data as text).



I can perform that test for you. From the data provided, I see two distinct Operator groups.

Hypothesis:

Null Hypothesis (H_0): There is no significant difference between the mean measurements of Operator A and Operator B.

Alternative Hypothesis (H_1): There is a significant difference.

Let's proceed with an independent samples t-test.

Results of Independent t-test:

t-statistic: 1.62

p-value: 0.11

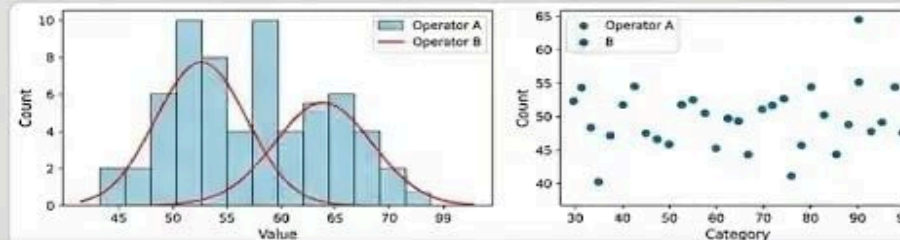
Sample sizes: $n_A=22$, $n_B=8$

Group Means: Mean_A=50.31, Mean_B=49.95

Conclusion:

With a p-value of 0.11 (which is greater than the standard alpha of 0.05), we fail to reject the null hypothesis. There is no significant difference between the operators.

To illustrate, here are the histograms and a text-based scatter plot.



Why R?

THE Statistical Language



R: The Statistical Language

